

SAT-Net: SincNet-based Attentive Temporal Convolutional Network for Motor Imagery Classification

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Abstract—Brain-computer interfaces (BCIs) are a promising method for users to interact with machines using brain signals, primarily electroencephalography (EEG). Motor imagery (MI)-BCI, which decodes EEG signals induced by the user's imagination of moving body parts, has gained great attention due to its applicability in various fields such as robotics and rehabilitation. MI-EEG signals exhibit class-discriminative patterns such as event-related de/synchronization across spectral-spatio-temporal (SST) domains. Numerous studies adopted deep learning framework, especially convolutional neural network (CNN), for learning SST feature representations automatically in a data-driven manner. In particular, most of the CNN-based methods adopted temporal convolution to learn the spectral information, as it can act as a band-pass filter. In this paper, we propose SAT-Net, a SincNet-based attentive temporal convolutional network for motor imagery classification. The proposed method utilizes Sinc convolution from SincNet for explicit extraction of spectral information from the input EEG signals with high interpretability. Moreover, we adopt an attentive temporal convolutional network to effectively learn SST feature representations while making full use of temporal information. We evaluate our proposed SAT-Net on the public BCI Competition IV-2a dataset, comparing it not only to conventional CNN-based approaches but also to the state-of-the-art method. The experimental results, supported by statistical analysis, demonstrate that our approach outperforms the competing methods.

Index Terms—Brain-computer Interface, Electroencephalography, Motor Imagery, SincNet, Temporal Convolutional Network

I. INTRODUCTION

Brain-computer interfaces (BCIs) are promising techniques that enable users to communicate with machines via brain signals, primarily electroencephalography (EEG) signals [1]. In particular, motor imagery (MI)-BCI is a system that controls

external devices by decoding EEG signals induced by the user's imagination of moving body parts [2]. It has attracted great attention due to its industrial and practical applicability in various fields such as robotics or rehabilitation [3]–[5].

MI-EEG signals exhibit discriminable patterns while carrying out different types of imagination (referred to as MI classes), such as Event-Related De/Synchronization (ERD/ERS) [6]. The differences in EEG patterns across different MI classes occur in the spectral, spatial, and temporal (SST) domains [7]. For example, when the user imagines moving their left hand, the ERD pattern usually emerges at the contralateral part of their brain, i.e., the right hemisphere, within the μ rhythm (8Hz–12Hz). Moreover, the patterns exhibit high variability across users in SST domains due to various factors such as gender, age, and concentration levels [8]. Therefore, numerous MI-BCI researchers have focused on extracting informative features from EEG signals in SST domains to implement an effective MI-BCI system.

Traditional MI-BCI studies adopted various methods for hand-crafted feature extraction in SST domains [9]–[12]. For instance, fast Fourier transform (FFT) and continuous wavelet transform (CWT) were commonly used for extracting spectral and spectral-temporal features of the EEG signals [13], [14]. Common spatial pattern (CSP) [15] was one of the most popular methods for learning class-discriminative spatial filters by maximizing differences in their variances between different MI classes to extract features. Notably, CSP-variants [9], [10] utilized filter bank to decompose the EEG signals into multiple sub-bands and learn the spatial filters for each sub-band. The classification results of these approaches were superior to the existing methods, including CSP, showing that the use of different frequency bands could enhance the classification

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performance.

Deep learning (DL)-based approaches have also been proposed, offering the advantage of learning informative features of EEG signals automatically in a data-driven manner [16]. Among various DL algorithms [17]–[19], CNNs have been widely used for EEG decoding due to their ability to learn high-level contextual and configurational information from input data without any prior knowledge [20]–[22]. For example, Lawhern et al. [23] proposed EEGNet, a compact CNN for EEG classification for various BCI paradigms including MI. EEGNet starts with temporal convolutional layer to learn frequency filters. Then, a depthwise convolution along the EEG channel-axis is employed to learn frequency-specific spatial filters. A separable convolution in the temporal domain is utilized as the last convolutional layer of EEGNet to fuse the extracted spectral-spatial features.

In particular, the majority of existing CNN-based methods including EEGNet adopted temporal convolutional layer as their first part of the network [23]–[25] to filter the EEG signals within different frequency bands. In detail, temporal convolutional kernel could be trained to act as a Finite-Impulse-Response (FIR) filter on the signals [26]. For example, in EEGNet, the length of the kernels was set to half of the sampling rate to capture spectral information at 2Hz and above. However, these approaches are implicit ways to extract spectral features since the spectral information that the output signals contain depends on the randomly initialized temporal kernels and the manually selected kernel length, which necessitate additional analyses for interpretation [27]. Although a few studies have attempted to alleviate this issue by utilizing multiple kernels with different lengths [24], manual selection of the kernel length could not always guarantee the best performance of MI classification.

In this regard, some of the latest studies proposed SincNet-based framework to explicitly learn the spectral information and provide high explainability [26], [28]. SincNet [29] was first proposed for a speaker recognition task, which used a sinc function as a convolutional kernel to act as band-pass filter based on two learnable parameters, low cut-off frequency and the filter bandwidth. By adopting the SincNet into MI classification task, the network was able to find the class-discriminative frequency bands automatically from the input EEG signals for each subject. Although SincNet-based frameworks in MI-BCI have achieved promising results with better interpretability compared to conventional CNN-based approaches, there is still room for further improvement.

Inspired by the previous works, we propose SAT-Net, a SincNet-based attentive temporal convolutional network for MI classification. Our method starts with sinc function-based convolutional (sinc-conv) layer, adopted from SincNet [29], to learn the class-discriminative spectral information from input EEG signals. Each kernel of the sinc-conv layer is initialized with different values of low cut-off frequency and bandwidth parameters, which acts like a learnable band-pass filter. Therefore, the sinc-conv layer outputs the EEG signals filtered in various class-discriminative frequency bands. The

filtered signals are fed into subsequent depthwise and temporal convolutional layers to learn SST feature representations. Since these layers output the temporal sequences of SST features, an additional process for extracting unexplored temporal information is required. Among the various time-series modeling algorithms, in this study, we propose a temporal convolutional network (TCN) [31] which efficiently learns high-level temporal information from the sequence. We apply multihead self-attention followed by TCN to first highlight the informative time steps from the sequence and extract high-level temporal features from SST feature sequences. To demonstrate the effectiveness of our proposed method, SAT-Net is evaluated on the public dataset, BCI competition IV-2a [32] with a statistical analysis.

The rest of the paper is organised as follows: The data description, details of the proposed SAT-Net, and experimental settings are demonstrated in Section II. Section III shows the experimental results and discussion.

II. MATERIAL AND METHODS

A. Dataset

In this paper, we used the BCI Competition IV-2a dataset [32] to train and evaluate our proposed method. The dataset consists of EEG signals from nine healthy subjects performing four different MI tasks (left hand, right hand, both feet, and tongue). For each subject, a total of two sessions were conducted on different days, where each session consisted of 72 trials for each task (total 288 trials). Each trial has EEG signals recorded from the 22 Ag/AgCl electrodes corresponding to the international 10-20 system [33] with the sampling rate of 250Hz. We cropped the EEG signals to a length of 4.5-second from 0.5 seconds before the start cue to 4 seconds after the start cue. The EEG signals were band-pass filtered between 0Hz to 40Hz, and exponential moving standardization was applied to the signals. In this paper, the first session was used for training and the second session is used for evaluation.

B. Learnable Band-pass Filtering based on Sinc Convolution

The sinc function-based convolutional (sinc-conv) layer [29] is utilized as the first layer of SAT-Net in order to learn class-discriminative spectral information from the input EEG signals $X \in \mathbb{R}^{C \times T_0}$, where C and T_0 represent the number of EEG electrodes and time points, respectively. Unlike a typical convolutional layer, the sinc-conv layer uses a parameterized sinc function as the convolutional kernel to implement a learnable band-pass filter on the input signals. Given a sinc function $\text{sinc}(x) = \sin(x)/x$, sinc convolution is defined as follows:

$$y[n] = x[n] * g[n, f_1, f_{bw}] = \sum_{k=0}^{k_0-1} x[k] * g[n-k, f_1, f_{bw}], \quad (1)$$

where $x[n]$ represents the input EEG signal in temporal domain and $y[n]$ refers to the signal filtered by the kernel

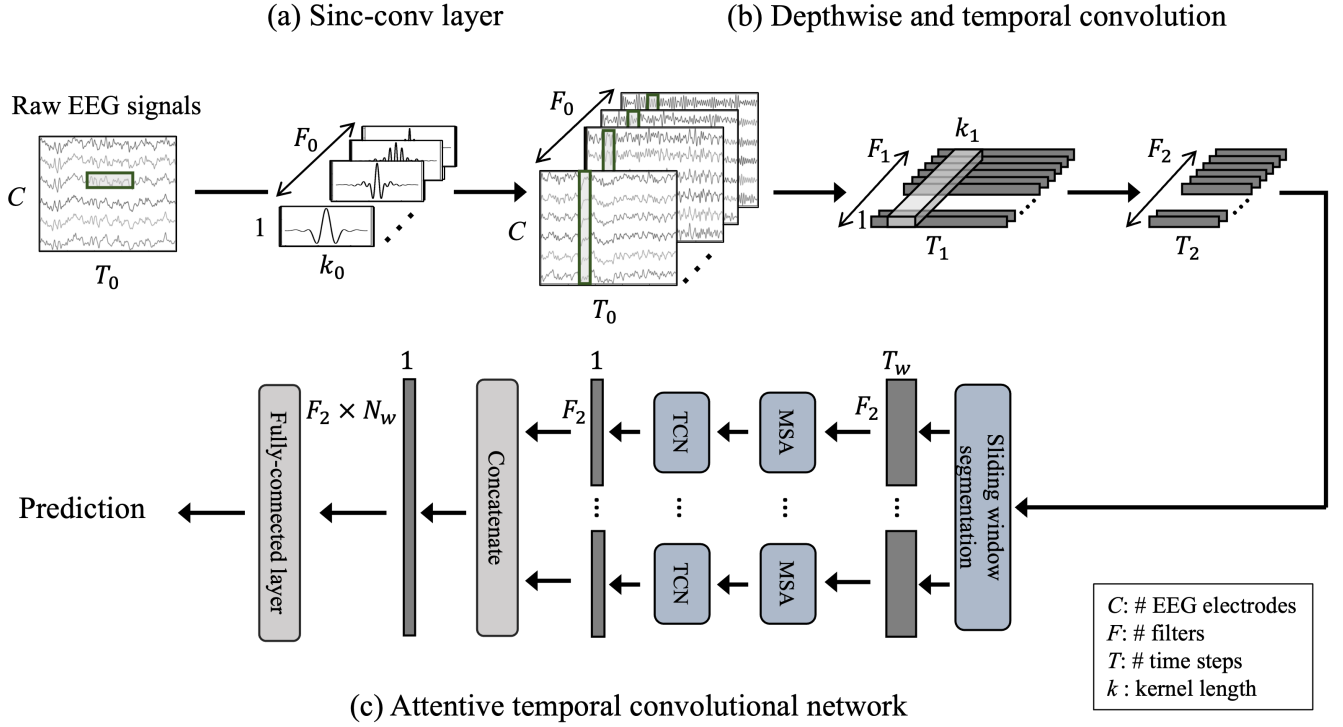


Fig. 1. Overview of the SAT-Net. (a) Sinc-conv layer to implement learnable band-pass filtering; (b) depthwise and temporal convolutional layers for SST feature extraction; and (c) attentive temporal convolutional network to further exploit temporal information. MSA: multihead self-attention [30], TCN: temporal convolutional network [31].

$g[\cdot]$ of length k_0 . The kernel g in the frequency domain can be written as the difference between two low-pass filters:

$$G[f, f_1, f_{bw}] = \text{rect}\left(\frac{f}{2f_2}\right) - \text{rect}\left(\frac{f}{2f_1}\right), \quad (2)$$

where $\text{rect}(\cdot)$ is a rectangular function, f_1 and f_{bw} are the learnable parameters denoting the low cut-off frequency and filter bandwidth, respectively, and $f_2 = f_1 + f_{bw}$ is high cut-off frequency. By using the inverse Fourier transform, the kernel g in the time domain can be written as:

$$g[n, f_1, f_{bw}] = 2f_2 \text{sinc}(2\pi f_2 n) - 2f_1 \text{sinc}(2\pi f_1 n). \quad (3)$$

Moreover, a Hamming window $w[\cdot]$ is applied to mitigate the issue of filter truncation [26] as:

$$g'[n, f_1, f_{bw}] = g[n, f_1, f_{bw}] * w[n], \quad (4)$$

where

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{k_0}\right). \quad (5)$$

In this paper, we exploit sinc-conv layer with F_0 kernels with different initial f_1 and f_{bw} parameters to filter the input EEG signals within from δ to γ rhythms (0Hz–36Hz), which are known to be important rhythms for MI classification [34]–[36]. For each kernel in sinc-conv layer, the parameters f_1 and f_{bw} are adjusted to selectively filter either low or high frequencies, and narrow or broad frequency bands, depending on the subject. That is, the sinc-conv layer is updated during

model training to capture class-discriminative frequency bands which may differ from subject-to-subject. Finally, the sinc-conv layers output the band-pass filtered EEG signals $H_0 \in \mathbb{R}^{F_0 \times C \times T_0}$.

C. Spatial Filtering and Temporal Convolution

The filtered EEG signals are fed into the depthwise convolutional layer to learn spatial information for different frequency bands. In detail, convolutional kernels with the shape of $(C, 1)$ is used for each of F_0 signals, which perform as frequency-specific spatial filters. The depth parameter D defines the number of spatial filters for each signals. Therefore, the output of depthwise convolution $H_d \in \mathbb{R}^{F_1 \times T_1}$ becomes the spectral-spatial features of EEG signals, where $F_1 = F_0 \times D$. Subsequently, temporal convolutional layer is adopted to fuse the extracted spectral-spatial features in the time domain. The temporal convolution operation outputs the summarized temporal sequence of SST features $H_t \in \mathbb{R}^{F_2 \times T_2}$, where there is still room for further extraction of temporal information.

D. Attentive Temporal Convolutional Network

To effectively and efficiently learn the available temporal information from extracted SST feature, H_t is segmented into N_w temporal sequences $H_w \in \mathbb{R}^{F_2 \times T_w}$ using sliding window of size T_w which can help to augment the features and improve the classification performance [20], [37]. In addition, for each SST segment H_w , multihead self-attention [30] is applied to

TABLE I
PERFORMANCE EVALUATION RESULTS OF OUR PROPOSED SAT-NET AND COMPETING METHODS ON THE BCI COMPETITION IV-2A DATASET.
(*: $p < 0.05$, **: $p < 0.001$)

Method	S01	S02	S03	S04	S05	S06	S07	S08	S09	Mean
DeepConvNet [20]	76.46 ± 1.38	46.87 ± 2.04	87.92 ± 1.46	67.78 ± 3.15	74.38 ± 1.76	56.53 ± 2.10	74.58 ± 2.14	77.22 ± 1.94	82.64 ± 1.61	71.60 $\pm 1.09^{**}$
ShallowConvNet [20]	76.04 ± 1.82	45.56 ± 5.66	85.07 ± 1.68	55.28 ± 2.65	62.29 ± 0.90	51.32 ± 1.71	69.58 ± 5.41	76.25 ± 1.19	74.79 ± 0.53	66.24 $\pm 1.31^{**}$
EEGNet [23]	77.29 ± 3.03	59.17 ± 2.39	88.47 ± 2.92	65.42 ± 1.88	71.18 ± 2.29	60.21 ± 2.02	72.71 ± 1.31	77.22 ± 1.88	76.32 ± 4.59	72.00 $\pm 0.51^{**}$
ATCNet [37]	78.31 ± 1.39	55.34 ± 0.55	84.39 ± 2.27	68.51 ± 0.99	72.97 ± 0.44	63.15 ± 2.49	85.50 ± 4.34	77.43 ± 4.82	81.25 ± 3.13	74.09 $\pm 0.62^*$
SAT-Net	79.98 ± 0.35	51.74 ± 0.08	91.20 ± 0.51	67.01 ± 0.65	74.54 ± 2.76	66.78 ± 0.99	89.00 ± 0.43	81.83 ± 0.83	83.22 ± 1.15	76.14 ± 0.01

re-calibrate the input SST feature. Then, each re-calibrated segment H'_w is fed into subsequent TCN [31] to further exploit temporal information by expanding the receptive field size to cover the entire sequence in temporal axis using dilated convolutions and residual blocks. Therefore, the output feature of TCN has an identical shape to the input H'_w , where each of T_w features with the dimension of F_2 contains SST information covering entire time steps of H'_w . The last element is selected from the output of TCN and concatenated with other output features from different windows and fed into the subsequent fully-connected layer for the prediction.

E. Experimental Settings

To train SAT-Net, we applied Adam optimizer with a learning rate of 4×10^{-3} to the cross-entropy loss function and used a cosine annealing scheduler for the convergence of the model training. We also used l_2 regularization with a coefficient of 2×10^{-3} to prevent overfitting. The mini-batch size was set to 72. For the sinc-conv layers, F_0 was set to 8 and the filter range of each kernel in the sinc-conv layer was set to 0Hz–8Hz, 4Hz–12Hz, $\dots$, 28Hz–36Hz. The length of the kernel k_0 in the sinc-conv layer was set to 125. The number of filters used in the depthwise convolution was set to 16 with the depth parameter D of 2. The number of temporal convolutional kernels F_2 was 64, where their length k_1 was set to 16. The size of the sliding window T_w was set to 16. We followed the detailed settings of multihead self-attention and TCN from the original paper [37]. We randomly selected 10% of the training set (session 1) for validation to determine the hyperparameters. Finally, we set the training epoch to 1500 to get the classification performance.

III. RESULT AND DISCUSSION

A. Experimental Results

In the study, we have conducted comparative experiments using the BCI Competition IV-2a dataset to evaluate our proposed SAT-Net. We have four baseline models for comparison: DeepConvNet [20], ShallowConvNet [20], EEGNet [23], and ATCNet [37], which are three conventional CNN-based

TABLE II
THE INITIAL AND UPDATED LOW CUT-OFF AND HIGH CUT-OFF FREQUENCIES OF TWO RANDOMLY SELECTED SUBJECTS. THE FILTER RANGE CONTAINING MU OR BETA RHYTHMS ARE EXPRESSED IN BOLD.

Sinc-conv kernel	Initial value		Subject 1		Subject 3	
	Low (Hz)	High (Hz)	Low (Hz)	High (Hz)	Low (Hz)	High (Hz)
1	0.0	8.0	0.0	4.0	0.0	4.0
2	4.0	12.0	10.81	36.60	7.42	11.42
3	8.0	16.0	8.95	26.49	9.97	32.45
4	12.0	20.0	9.70	30.30	9.42	33.84
5	16.0	24.0	18.13	33.76	16.20	20.20
6	20.0	28.0	16.42	28.44	16.62	20.62
7	24.0	32.0	24.78	36.97	17.05	21.05
8	28.0	36.0	30.19	38.59	29.42	33.42

models, and one state-of-the-art (SOTA) model, respectively. Moreover, all experiments were performed five times using different random seeds each time, then the accuracies for each method and subject were averaged for a thorough evaluation. We have also employed the Wilcoxon signed-rank test [38] to further validate the significance of our SAT-Net and assess the differences between the baseline models.

The experimental results demonstrated that our proposed SAT-Net achieved the highest mean accuracy compared to all the baseline models, with its performance improvement up to 9.54%, as shown in Table I. Our method even exhibited a performance improvement of 1.69% over the SOTA model, with a smaller variance as well. Upon the results for each individual subject, our model outperformed the competing methods in the majority of subjects. In detail, the statistical analysis also demonstrated that our model significantly outperformed competing methods with $p < 0.05$ for ATCNet and $p < 0.001$ for others.

B. Discussion

1) *Neurophysiological Analysis on Sinc-conv Layer:* In Table I, our proposed SAT-Net outperformed the state-of-the-art method, ATCNet [37], in seven out of the nine subjects by using the sinc-conv layer instead of the conventional temporal convolutional layer. Even though the temporal convolution in ATCNet perform as a band-pass filter, the extracted spectral information depends on the manually selected length of the kernel and randomly initialized parameters, which cannot guarantee the quality of extracted spectral features and degrade the model interpretability. In contrast, in this study, the sinc-conv layer is initialized to filter within the frequency band that is generally known to be effective for MI classification (δ to γ rhythm). Then, as shown in Table II, it is optimized to learn the informative frequency band in a subject-specific manner, which also provides high interpretability based on its low cut-off frequency and filter bandwidth parameters. Table II shows the initial and updated filter ranges of the sinc-conv layers for two randomly selected subjects. The initial filter ranges were set to filter the EEG signals within 0Hz to 36Hz with a bandwidth of 8Hz. After training, most of the sinc-conv layers for both subjects were updated to filter the EEG signals into various frequency ranges, especially capturing μ to β rhythm (8Hz–30Hz). Of note, the low cut-off frequency and filter bandwidth parameters after training varied across the two subjects. For example, the filters of Subject 1 captured the spectral information within not only $\mu - \beta$ rhythm but also the high frequency range, i.e., the γ rhythm, while those of Subject 3 captured only the frequency range under β rhythm. This indicates that each subject has different optimal frequency ranges and coupling combinations.

2) *Limitations and Future Works:* Although SAT-Net demonstrated improved performance for the majority of individual subjects and the average accuracy, a notable decrease was observed in the performance of Subject 2. The subject was considered to exhibit BCI-illiteracy in prior MI studies, indicating difficulties in imagining body movements [39]. This resulted in the network making incorrect decisions, and as a consequence, the sinc-conv layer was unable to efficiently optimize and capture informative frequencies, leading to a degradation in performance. Moreover, our experiments were limited to a single dataset comprising only nine subjects, where large MI datasets have been released to the public in recent years. In our future research, we aim to enhance MI classification performance for individuals with BCI-illiteracy by leveraging information from other subjects on various datasets to validate the generalizability and superiority of the proposed method [40].

IV. CONCLUSION

In this paper, we proposed SAT-Net, a SincNet-based attentive temporal convolutional network for MI classification. By utilizing sinc-conv layer, SAT-Net filter the input EEG signals into class-discriminative frequency bands with high interpretability. Subsequently, the depwthwise and temporal convolution extracted and merged the spectral-spatial features,

generating the sequence of SST features. Finally, an attentive temporal convolutional network was employed to further exploit the remaining temporal information from the SST feature sequences. The experimental results demonstrated that our proposed SAT-Net significantly outperformed not only the conventional methods but also the state-of-the-art (SOTA) method.

ACKNOWLEDGMENT

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No. 2019-0-00079, Artificial Intelligence Graduate School Program(Korea University), No. 2017-0-00451, Development of BCI based Brain and Cognitive Computing Technology for Recognizing User's Intentions using Deep Learning, No. 2022-0-00871, Development of AI Autonomy and Knowledge Enhancement for AI Agent Collaboration), and the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS202300212498)

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